Chaos testing in Restauranty App

An early report - Chaos Mesh

1. **Pod Fault > Pod Kill**

Goal: Recovery after pods were killed off

Applied Test:

```yaml
kind: PodChaos
apiVersion: chaos-mesh.org/v1alpha1
metadata:
  namespace: restauranty
  name: node-failure-simulation
spec:
  selector:
    namespaces:
      - restauranty
    pods:
      restauranty:
        - auth-deployment-7d8f4c774-b5rh9
        - discounts-deployment-74d66fdc76-rlns6
        - items-deployment-66bf65556-254z4
  mode: all
  action: pod-kill
```

Pods difference

```
jal98@PC2024Jalmeida:~/Project3/restauranty-public$ kubectl get pods -n restauranty -o wide -w
auth-deployment-7d8f4c774-b5rh9         1/1   Terminating   3 (72m ago)    78m   10.244.7.10    aks-default-25801085-vmss00000p   <none>   <none>
auth-deployment-7d8f4c774-b5rh9         1/1   Terminating   3 (72m ago)    78m   10.244.7.10    aks-default-25801085-vmss00000p   <none>   <none>
discounts-deployment-74d66fdc76-rlns6   1/1   Terminating   0              26h   10.244.7.173   aks-default-25801085-vmss00000p   <none>   <none>
discounts-deployment-74d66fdc76-rlns6   1/1   Terminating   0              26h   10.244.7.173   aks-default-25801085-vmss00000p   <none>   <none>
items-deployment-66bf65556-254z4        1/1   Terminating   4 (4h15m ago)  26h   10.244.7.93    aks-default-25801085-vmss00000p   <none>   <none>
auth-deployment-7d8f4c774-g8jgk         0/1   Pending       0              0s    <none>         <none>                            <none>   <none>
discounts-deployment-74d66fdc76-h25d9   0/1   Pending       0              0s    <none>         <none>                            <none>   <none>
auth-deployment-7d8f4c774-g8jgk         0/1   Pending       0              0s    <none>         aks-bigger-32857848-vmss000000    <none>   <none>
items-deployment-66bf65556-254z4        1/1   Terminating   4 (4h15m ago)  26h   10.244.7.93    aks-default-25801085-vmss00000p   <none>   <none>
discounts-deployment-74d66fdc76-h25d9   0/1   Pending       0              0s    <none>         aks-default-25801085-vmss00000p   <none>   <none>
items-deployment-66bf65556-pm2bm        0/1   Pending       0              0s    <none>         <none>                            <none>   <none>
items-deployment-66bf65556-pm2bm        0/1   Pending       0              0s    <none>         aks-default-25801085-vmss00000p   <none>   <none>
auth-deployment-7d8f4c774-g8jgk         0/1   ContainerCreating   0          0s      <none>         aks-bigger-32857848-vmss000000    <none>   <none>
auth-deployment-7d8f4c774-g8jgk         0/1   Running             0          2s      10.244.2.50    aks-bigger-32857848-vmss000000    <none>   <none>
discounts-deployment-74d66fdc76-h25d9   0/1   ContainerCreating   0          8s      <none>         aks-default-25801085-vmss00000p   <none>   <none>
items-deployment-66bf65556-pm2bm        0/1   ContainerCreating   0          9s      <none>         aks-default-25801085-vmss00000p   <none>   <none>
auth-deployment-7d8f4c774-g8jgk         1/1   Running             0          13s     10.244.2.50    aks-bigger-32857848-vmss000000    <none>   <none>
discounts-deployment-74d66fdc76-h25d9   0/1   Running             0          58s     10.244.7.187   aks-default-25801085-vmss00000p   <none>   <none>
items-deployment-66bf65556-pm2bm        0/1   Running             0          70s     10.244.7.231   aks-default-25801085-vmss00000p   <none>   <none>
discounts-deployment-74d66fdc76-h25d9   1/1   Running             0          2m30s   10.244.7.187   aks-default-25801085-vmss00000p   <none>   <none>
items-deployment-66bf65556-pm2bm        1/1   Running             0          2m31s   10.244.7.231   aks-default-25801085-vmss00000p   <none>   <none>
```

Conclusion: After 3 pods were killed, kubernetes recreated the pods

Result: OK - Success

2. Stress Testing

Goal: Ability to scale up and down properly by using HPA in auth deployment (Horizontal Pod Autoscaller)

Applied Test:

```yaml
kind: StressChaos
apiVersion: chaos-mesh.org/v1alpha1
metadata:
  namespace: restauranty
  name: stress-testing-auth
spec:
  selector:
    namespaces:
      - restauranty
    labelSelectors:
      app: auth
  mode: all
  stressors:
    cpu:
      workers: 2
      load: 100
  duration: 5m
```

Results:

auth-hpa	cpu: 7%/60%	REPLICAS: 2
auth-hpa	cpu: 105%/60%	REPLICAS: 2
auth-hpa	cpu: 105%/60%	REPLICAS: 4
auth-hpa	cpu: 250%/60%	REPLICAS: 4
auth-hpa	cpu: 250%/60%	REPLICAS: 5
auth-hpa	cpu: 168%/60%	REPLICAS: 5
auth-hpa	cpu: 132%/60%	REPLICAS: 5
auth-hpa	cpu: 168%/60%	REPLICAS: 5
auth-hpa	cpu: 70%/60%	REPLICAS: 5
auth-hpa	cpu: 54%/60%	REPLICAS: 5
auth-hpa	cpu: 12%/60%	REPLICAS: 5
auth-hpa	cpu: 5%/60%	REPLICAS: 5
auth-hpa	cpu: 12%/60%	REPLICAS: 5
auth-hpa	cpu: 5%/60%	REPLICAS: 5
auth-hpa	cpu: 12%/60%	REPLICAS: 5
auth-hpa	cpu: 9%/60%	REPLICAS: 5
auth-hpa	cpu: 9%/60%	REPLICAS: 5
auth-hpa	cpu: 8%/60%	REPLICAS: 4
auth-hpa	cpu: 9%/60%	REPLICAS: 4
auth-hpa	cpu: 5%/60%	REPLICAS: 4
auth-hpa	cpu: 8%/60%	REPLICAS: 4
auth-hpa	cpu: 9%/60%	REPLICAS: 4
auth-hpa	cpu: 15%/60%	REPLICAS: 2
auth-hpa	cpu: 13%/60%	REPLICAS: 2
auth-hpa	cpu: 15%/60%	REPLICAS: 2

Notes: Quick replica creation, but slower replica reduction (300s) to avoid fluctuations

```
    scaleDown:
      stabilizationWindowSeconds: 300
    scaleUp:
      stabilizationWindowSeconds: 0
```

Conclusion: HPA tested with sucess
Result: OK - Success

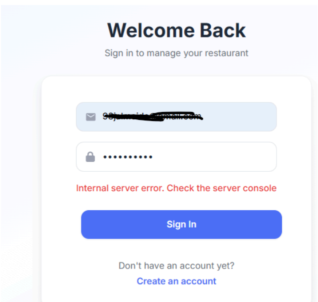
3. Network Chaos

Goal: Simulate DB latency

Test applied:

```yaml
kind: NetworkChaos
apiVersion: chaos-mesh.org/v1alpha1
metadata:
  namespace: restauranty
  name: mongo-latency
spec:
  selector:
    namespaces:
      - restauranty
    labelSelectors:
      app: auth
  mode: all
  action: partition
  duration: 2m
  direction: to
  target:
    selector:
      namespaces:
        - restauranty
      labelSelectors:
        app: my-mongo
    mode: all
```

Results:
During the test, it was unable to authenticate in the restauranty-app, as detailed bellow. The logs proved a timeout when connecting to mongodb, resulting in a 5xx error while the auth pods were running.



Start time	Trace Service	Trace Name	Duration
2026-09-09 22:41:27	auth-service	countDocuments users	1.27 mins
2026-09-09 22:42:0	auth-service	countDocuments users	43.6 s
2026-09-09 22:42:3	auth-service	POST /api/auth/login	30.6 s
2026-09-09 22:42:5	auth-service	POST /api/auth/login	30.2 s
2026-09-09 22:42:3	auth-service	POST /api/auth/login	30.2 s
2026-09-09 22:42:1	auth-service	tcp.connect	30.2 s
2026-09-09 22:42:3	auth-service	POST /api/auth/login	30.1 s
2026-09-09 22:42:3	auth-service	POST /api/auth/login	30.1 s
2026-09-09 22:42:4	auth-service	POST /api/auth/login	30.1 s
2026-09-09 22:42:3	auth-service	POST /api/auth/login	30.1 s
2026-09-09 22:42:4	auth-service	tcp.connect	30.0 s
2026-09-09 22:42:2	auth-service	countDocuments users	30.0 s

```
jal98@PC2024Jalmeida:~/Project3/restauranty-public$ kubectl exec -it deploy/auth-deployment -n restauranty -- sh
/app # wget -qO- --timeout=5 http://my-mongo.restauranty.svc.cluster.local:27017
wget: download timed out
/app # wget -qO- --timeout=5 http://my-mongo.restauranty.svc.cluster.local:27017
wget: download timed out
/app # wget -qO- --timeout=5 http://my-mongo.restauranty.svc.cluster.local:27017
wget: download timed out
/app # wget -qO- --timeout=5 http://my-mongo.restauranty.svc.cluster.local:27017
It looks like you are trying to access MongoDB over HTTP on the native driver port.
/app # wget -qO- --timeout=5 http://my-mongo.restauranty.svc.cluster.local:27017
It looks like you are trying to access MongoDB over HTTP on the native driver port.
/app #
```

Conclusion: After review, the correct behaviour would be for the pods affected by a crashloop scenario to enhance the visibility of the occurrence. The same behaviour was also tested and identified in items and discounts.
Suggestion: Develop an endpoint that the service can use as a readiness probe .
Result: NOK - Unsuccessful